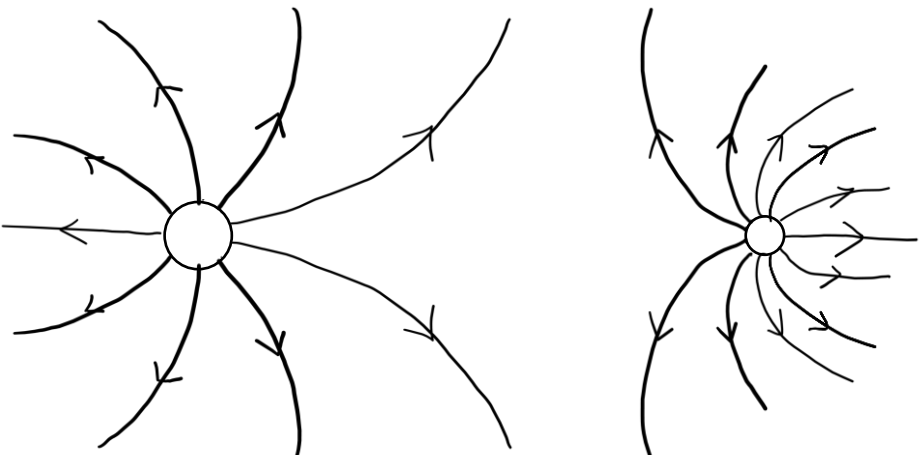


4a	Zero electric field strengths in sphere A (between $x = 0$ and $x = 1.4$ cm) and in sphere B (between $x = 11.4$ and $x = 12.0$ cm)	B1
4b	Since the <u>field strength is zero at a point between the spheres</u> OR the <u>electric fields are in opposite directions</u> , the charges on the spheres are of the <u>same sign</u> .	M1 A1
4c	At $x = 0.08$ m, the electric field strength due to sphere A cancels out the electric field strength due to sphere B. $E_A = E_B$ $\frac{Q_A}{4\pi\epsilon_0 (0.08)^2} = \frac{Q_B}{4\pi\epsilon_0 (0.04)^2}$ $\frac{Q_A}{Q_B} = \left(\frac{0.08}{0.04}\right)^2$ $= 4$ (Allow estimation from graph, $7.8 \text{ cm} < x < 8.2 \text{ cm}$)	B1 C1 A1
4d	 • Correct field line direction (either all inwards or all outwards) and shape • neutral point nearer to sphere B and any of the following: • field lines more closely spaced for sphere B • field lines perpendicularly from the spheres	B1 B1 B1
4e	change in electric potential = area under graph between $x = 1.4$ cm and $x = 3.0$ cm Estimated area between $x = 1.4$ cm and $x = 3.0$ cm = $\frac{1}{2} (0.8 \times 10^{-2})(11.0 + 5.0)10^7 + \frac{1}{2} (0.8 \times 10^{-2})(5.0 + 2.5)10^7$ = 940 kV Or by counting squares under graph. Acceptable range for either method is between 850 kV and 1050 kV.	C1 C1

	Energy gained by proton = $(940 \times 10^3)(1.60 \times 10^{-19})$ $= 1.5 \times 10^{-13} \text{ J}$ Accept $(1.4 \text{ to } 1.6) \times 10^{-13} \text{ J}$	
	Total:	A1 12

52(i)		
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